Tiny Pond Creature Breaks Usual DNA "Stop" Rules

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Scientists testing a **DNA** tool found a surprise in a tiny pond creature. In most living things, certain **DNA** signals tell a cell to stop building a **protein**. In this **protist**, those "stop" signals instead code for two different **amino acids**the building blocks of **proteins**.

Dr. Jamie McGowan studied the organism from a freshwater pond at Oxford University Parks in England. The team was not hunting for weird codes. They were testing a way to read **DNA** from very small samples. When they looked at the **genome**, the rules looked different.

The **protist** is called Oligohymenophorea sp. PL0344. Findings in PLOS Genetics showed the genetic code can be more flexible than many people thought. Later work found more odd codes in related microbes.

Protists are living things with cell nuclei that are not animals, plants, or fungi. McGowan said it was lucky they picked this oneand it shows how much is still unknown.

Word Treasure

protist -- A tiny living thing that is not an animal, plant, or fungus.
DNA -- The chemical code inside cells that carries instructions for life.
amino acids -- Small parts that link together to build proteins.
protein -- A molecule cells build to do important jobs in the body.
genome -- All of an organism's genetic instructions.

Background you need

DNA holds the instructions cells use to build proteins.

Special DNA signals usually tell a cell when to stop making a protein.

Protists are tiny living things that scientists are still learning about.

Break it down

WHO: Scientists including Dr. Jamie McGowan
WHAT: Finding a protist that uses DNA stop signals in a new way
WHERE: A pond at Oxford University Parks in England
WHY: The strange code showed life's instructions can have rare exceptions

Why it matters

Science surprises remind kids that nature still has mysteries to discover.

Test what you remember

1. Where was the tiny organism found?

- (A) A desert oven
- (B) The Moon
- (C) Inside a soccer ball
- (D) A freshwater pond in England

2. What do stop signals usually do?

- (A) End a soccer game
- (B) Tell a cell to stop making a protein
- (C) Stop rain
- (D) Turn off streetlights

3. What were the signals changed to code for?

- (A) Two sports
- (B) Two different amino acids
- (C) Two planets
- (D) Two songs

4. Who helped make the discovery?

- (A) A pilot
- (B) A chef
- (C) A weather reporter
- (D) Dr. Jamie McGowan

5. What is a protist in this story?

- (A) A musical note
- (B) A type of truck
- (C) A living thing that is not an animal, plant, or fungus
- (D) A sports rule

6. Where were findings published?

- (A) A sports blog
- (B) PLOS Genetics
- (C) A comic only
- (D) A cookbook

Answer key (try the quiz first!): 1: D 2: B 3: B 4: D 5: C 6: B

Questions to think about

1. Dr. Jamie McGowan: Says choosing this protist was lucky and shows how little we know.

MY THOUGHTS (at least 20 words):